EVALUATING LIMITS

Factoring and canceling

When substitution gives $0/0$, numerator and denominator share a factor. Factor both, cancel it, then substitute.

01

SUBSTITUTION GIVES 0/0

BEFORE

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$$

↓ Factor the numerator — $x^2 - 4 = (x - 2)(x + 2)$

02

SHARED FACTOR VISIBLE

$$\lim_{x \to 2} \frac{(x - 2)(x + 2)}{x - 2}$$

↓ Cancel (x - 2) — allowed: for x near 2 but not at 2 the factor is nonzero

03

SUBSTITUTE

AFTER

$$\lim_{x \to 2}(x + 2) = 4$$

▼ CANCEL FACTORS, NOT TERMS